CURBCOUNT · DESIGN DECISIONS REGISTER

CurbCount design decisions register

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	CBC-DEC-001	Version	0.5
Status	Draft	Date	2026-10-31
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/curbcount		
Source	docs/06-design-decisions.md @ 82b4850		

REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-31	Amish Chadha	Kit 1.9.1 upgrade: open decisions O1 (cold charging) and O2 (concept choice) added; C1 (radar duty cycle) and C2 (model rebuilt, no dimension changed) recorded as decided under the standing instruction
0.4	2026-10-02	Amish Chadha	Decisions 5 to 9 carried into model.py, the appearance model and the BOM
0.3	2026-10-02	Amish Chadha	Amish approved the recommendations for all nine open decisions on 2026-10-02 (CBC-DDR-003 accepted); moved to decisions made
0.2	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.1	2026-09-30	Amish Chadha	Register opened with the build plan; open decisions gathered from the review note and the decision records

CurbCount design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` or in the review note; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`, CBC-BLD-001) describes the design as it stands and does not list open decisions.

Open decisions

Table 1. Open decisions.

#	Decision needed	Options	Recommendation	Affects in the build	Source
O1	Cold charging: LiFePO4 may not be charged below 0 degrees C, so charging stops below about 0 degrees C air (STM S8)	(1) limit the claim to cool sites; (2) heated cell pocket in the FieldNode core; (3) low-temperature cell	Option 1 now (done in PR-010), and raise 2 and 3 with the FieldNode project	BOM line 3 and the FieldNode enclosure if 2 or 3; PR-010; CAL-001 E	CBC-DDR-004, O1; CBC-STM-001 S8
O2	Concept choice between the thermal array head (A) and the 60 GHz radar head (B); scores 121 and 110, within 10 % so no pick is made	(1) A as the build, B as the hot-climate variant; (2) B for all sites; (3) a combined head	Option 1 (already decided on 2026-09-25; nothing changes)	Nothing if 1; the whole head, panel and cells if 2	CBC-DDR-004, O2; CBC-OPT-001
O3	Vehicle counting: a car's warm hood is in the field for 3.2 frames, not 4 (STM S5)	(1) 32 Hz subpages with 1 MHz I2C: 6.4 frames, noise up by the square root of 2, a faster pull-up; (2) accept that vehicle counts are not met and say so; (3) a vehicle sensor on option B only	Option 1: it is firmware and bus speed, but it sets the shared FieldNode sensor port speed, so FieldNode must agree	CAL-001 C, F; PR-001, PR-003; FieldNode sensor port	CBC-DDR-004, O3; CBC-STM-001 S5
O4	If option B is carried forward: FieldNode's high-load charger for the 20 W panel (5.0 A against the 1.5 A setting, STM S28) and a shield or shaded cell for hot sites (STM S26)	(1) FieldNode high-load variant plus shield; (2) leave B as a thermal-only-hot-site study	Option 1, argued in the FieldNode project	Option B's BOM, enclosure and panel; nothing in option A	CBC-DDR-004, O4; CBC-STM-001 S26, S28

To confirm when parts are bought

Table 2. Things to check against the real part before it is used.

#	What to confirm	Why it matters	Source
1	The solar panel's frame has a flat back lip at least 12 mm wide at its short ends	The rail plate bolts through it; without it the panel cannot be fixed as drawn	CBC-DDR-003, P5
2	The enclosure model, its boss spacing (110 x 160 mm assumed) and its lug kit	They set the internal plate holes and the lug screw holes in the saddle plate	FND-DDR-003; CBC-DDR-003, P2
3	Band clamps that close on the loop for the site pole: about 470 mm on a 114 mm pole (330 mm on 60 mm, 545 mm on 140 mm)	A worm-drive band has a limited range; the wrong size will not tighten	CBC-DDR-003, P1
4	The sleeve tube's bore is at least 122 mm, and the cap disc fits it closely	The sleeve must clear a 114.3 mm pole by 3 to 4 mm all round	CBC-DDR-003, P5
5	The band torque that gives 1,000 N of preload	The twist calculation assumes that preload	CBC-CAL-001 [H3]
6	The sensor cable has five cores of about 0.25 mm², a moulded M12 A-coded plug and an open end	The open end goes through the head gland to the breakout	CBC-DDR-003, P6
7	Prices of every BOM line	The estimate is $1.00 under the $275 value-engineering target	CBC-CAL-001 [J1]

Value engineering

Value-engineering target: USD 275 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 274.00 (USD 1.00 under the target). The estimate rests on indicative prices.

Main cost drivers: the FieldNode share (lines 1 to 3, 14 and part of line 13) at USD 105.00, FieldNode's own panel (line 4) at USD 14, and the parts added for construction together with FieldNode's repricing of its core, which took the total from USD 253.00 to USD 274.00. The ESP32-S3 fallback head would cost about USD 311.00, over the target.

Savings worth trying: leaving port B out of CurbCount's core (about USD 6 less, departing from the standard FieldNode core), and confirming prices when parts are bought.

Decisions made

Table 3. Decisions made, with the words recorded.

Date	Decision	Decided by	Record
2026-09-25	Batch taken through TRL 3, nothing past it	Amish: "you know the drill, nothing gets past TRL 3"	REVIEW 2026-09-25 (TRL 3)
2026-09-25	TRL 2 items D1 to D7: $275 value-engineering target, thermal array first with radar as the hot-climate variant, FieldNode high-load variant raised with FieldNode, processor study, 15-minute bins and three classes, jumper-only calibration, public notice plate	Amish: "i accept all your recommendations, go with them across all repos."	CBC-DDR-001, CBC-DDR-002
2026-09-25	Tracking on FieldNode's STM32WL as the baseline with the ESP32-S3 head as fallback; one cell and the 6 W panel; R4 restated in px/m; R7 for temperate sites; R11 kept at 6 kg; 10-byte record	Amish, same instruction	CBC-DDR-002, items 3 to 8
2026-09-30	Build plan format approved for all repos; outstanding decisions kept out of the build plan, in this register	Amish: "this is the correct build plan ... this is a good quality document format. Extend this across all the other repos"	This register; CBC-BLD-001
2026-09-30	Make the design physically buildable while drawing the build plan	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	CBC-DDR-003 (changes made under this instruction; accepted on 2026-10-02, below)
2026-10-02	Design for construction accepted as a whole: the changes P1 to P7 of CBC-DDR-003 and their knock-on changes, as made	Amish: "i approve your recommendations for all 555 open decisions."	CBC-DDR-003, A2
2026-10-02	First co-design partner: a city transport department that already runs manual or loop counts on a street with a bike lane and poles of 60 to 140 mm, with the first trial in a season with air under 20 °C. First candidate to approach: the City of Toronto's transportation services, which publishes its traffic count data	Amish: "i approve your recommendations for all 555 open decisions."	CBC-DDR-002, O1
2026-10-02	Sensor port: CurbCount's position to agree with FieldNode is one common pin order for both ports (supply, ground, I2C data, I2C clock, spare), each port's supply voltage set by a per-project supply module (switched 3.3 V on port A for CurbCount), and the calibration jumper sensed on FieldNode's service header	Amish: "i approve your recommendations for all 555 open decisions."	REVIEW 2026-09-25 cross-repo actions; FND-DDR-001, O2
2026-10-02	Pole top drawn 560 mm lower in the photoreal renders accepted as a render-only layout; the caption says it is drawn closer than installed	Amish: "i approve your recommendations for all 555 open decisions."	REVIEW 2026-09-26, item 1
2026-10-02			

Date	Decision	Decided by	Record
	Notice plate rendered at its installed 2.6 m, not 4.15 m	Amish: "i approve your recommendations for all 555 open decisions."	REVIEW 2026-09-26, item 2
2026-10-02	Notice plate wording adopted with one fix: "No images are stored or sent; only counts leave the device" replaces "No images are taken or stored"; the rest and the repository link are kept	Amish: "i approve your recommendations for all 555 open decisions."	REVIEW 2026-09-26, item 3
2026-10-02	Status light on the sensor head dropped	Amish: "i approve your recommendations for all 555 open decisions."	REVIEW 2026-09-26, item 4
2026-10-02	"COUNTS ONLY" plaque embossed into the printed head housing (BOM line 8); the enclosure label is part of BOM line 1	Amish: "i approve your recommendations for all 555 open decisions."	REVIEW 2026-09-26, item 5
2026-10-02	Head window bezel and head bolts of the appearance model closed as superseded by the window frame and M5 bolts of CBC-DDR-003, P4; the appearance model was updated to match on 2026-10-02	Amish: "i approve your recommendations for all 555 open decisions."	REVIEW 2026-09-26, item 6; CBC-DDR-003, P4
2026-10-31	C1: option B's radar runs in frame bursts for about 0.2 W average, one burst in six (was 1.2 W continuous, which drew 32 Wh/day against 14.5 Wh/day stored; 0.25 W still gave only 2.6 days with cold and end of life together); two cells and a 20 W panel unchanged	Claude, under Amish's standing instruction of 2026-10-23 (fix rule); Amish can reverse it	CBC-DDR-004, C1; STM S22, S23
2026-10-31	C2: model.py rebuilt to the kit 1.9 interface at D3 for both options with every fixing as a part; `PARAMS` and `derived()` identical to the kit 1.7 model, so no design dimension changed	Claude, under the same instruction (modelling only)	CBC-DDR-004, C2; cad/fidelity.json